Aniket Pratap

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Education

University of California San Diego
Master of Computer Science — *Expected Dec 2025*

University of California Irvine — 3.83 GPA
Bachelor of Computer Science — *Graduated Jun 2024*

Experience

Algorithms Intern — June 2025 - Sept 2025
KLA - MACH Team — *Ann Arbor, MI*
- Developed a high-performance physics simulation using CUDA to model wafer etching processes
- Implemented parallel particle interactions using CUDA threads, enabling concurrent etching computations
- Accelerated compute pipeline by 100× through GPU parallelization, cutting client runtime from hours to minutes

Software Engineering Intern — Apr 2024 - Sept 2024
Ekastar — *Remote*
- Developed features for Ekastar's Order Management System, which supported daily client transactions and improved platform stability
- Built dashboards in React/TypeScript to track product data and delivery, reducing manual reporting by 30%
- Implemented secure .NET API authentication, scaled reliably to 10,000+ accounts with reduced access errors

NLP Researcher — June 2023 - Apr 2024
University of California Irvine — *Irvine, CA*
- Architected a machine learning model using live scraped data to analyze user intention, which governed a bot to facilitate chat conversations
- Fine-tuned GPT2 using PyTorch and user data to achieve an accuracy of 98% when analyzing user sentiment, leading to refined bot messages and user engagement in focus groups
- Implemented NumPy and vectorization to speed model training by 40%

Software Engineering Intern — June 2022 – Aug 2022
Cordis — *San Diego, CA*
- Automated sales mapping across 800 territories, eliminating a week of manual input and boosting team productivity
- Designed a web training program to onboard 1000+ business analysts, saving $10K by replacing third-party solutions

Projects

ZotClass | *React Native, Android, iOS, ZotAPI* — https://github.com/xXViridianXx/ZotClass
- Developed a full-stack React Native super-app enabling UCI students to explore available courses effortlessly
- Implemented ZotAPI to retrieve real-time class status, descriptions, and prerequisites to enhance user experience
- Scaled app to serve 1000+ active users

ZotAPI | *Python, Beautiful Soup, AWS, Lambda, DynamoDB* — https://github.com/xXViridianXx/Zot-API
- Designed a Python web scraper to extract, parse, and clean data from university course listings, forming the backbone of a robust API
- Improved scraping performance by 99% using AWS Lambda for distributed execution, ensuring faster data processing and storage in DynamoDB for efficient querying

Technical Skills

Programming: Python, C/C++, C#, CUDA, SQL, JavaScript, Swift
ML/AI: PyTorch, TensorFlow, NumPy, Pandas
Web/Full-Stack: React, React Native, Node.js, Flask, Django, Tailwind, Redux
Cloud/Tools: AWS, Docker, Git, Firebase, Tableau, Oracle Analytics, Figma